

On the Expressive Power and Generalization Limits of Quantum Data Reuploaders

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- Variational Quantum Circuits (VQCs) are central to near-term QML
- **Data reuploading** is widely used to enhance expressivity
- Empirically effective — but:
 - Why does reuploading help?
 - How does expressivity scale with depth?
 - Does expressivity saturate?
- This work provides a **theoretical explanation** using Fourier analysis

Quantum Data Reuploaders (QDRs)

A general R-reupload quantum neural network:

$$U(\boldsymbol{\theta}, \boldsymbol{x}) = W_R V(\boldsymbol{x}) W_{R-1} \cdots V(\boldsymbol{x}) W_1$$

- W_i : trainable unitaries
- $V(\boldsymbol{x}) = e^{-i \sum_j x_j H_j}$: data encoding
- Model output:

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \langle 0^{\otimes n} | U^\dagger M U | 0^{\otimes n} \rangle$$

Key Idea: Fourier View of QDRs

- Data enters the circuit via exponentials of Hermitian operators
- Such models compute **trigonometric polynomials**
- Following Schuld (2021):
 - QDR outputs admit a **finite Fourier expansion**
 - Expressivity = accessible frequency spectrum

Main Result:

$$f_{\theta}(x) = \sum_{k \in K_R} c_k(\theta) e^{i k \cdot x}$$

- K_R : frequency set determined by encoding Hamiltonians
- Frequencies arise from eigenvalues of $\{H_j\}$
- Each reupload **adds** spectral components

Origin of Frequencies in QDRs

Each data-encoding unitary has the form

$$V(x) = e^{-ixH}$$

- H has finite spectrum $\sigma(H) = \{\lambda\}$
- Spectral decomposition gives:

$$e^{-ixH} = \sum_{\lambda \in \sigma(H)} e^{-ix\lambda} \Pi_{\lambda}$$

- Each encoding contributes a **phase factor** $e^{-ix\lambda}$

Key idea: Eigenvalues of H directly determine Fourier frequencies.

Effect of Reuploading

For an R -reupload circuit:

$$U(x) = V(x)W_R \cdots V(x)W_1$$

- Each encoding contributes one eigenvalue
- A typical term accumulates phase:

$$e^{-ix(\lambda_1 + \lambda_2 + \cdots + \lambda_R)}$$

- Hence all accumulated phases belong to:

$$\alpha \in S^{(R)} := \underbrace{S + \cdots + S}_{R \text{ times}}$$

Conclusion: Every term in $U(x)$ carries a frequency from $S^{(R)}$.

Why Frequencies Are Differences

The model output is:

$$f(x) = \langle \psi | U^\dagger(x) O U(x) | \psi \rangle$$

- $U(x)$ contributes $e^{-ix\alpha}$, $\alpha \in S^{(R)}$
- $U^\dagger(x)$ contributes $e^{+ix\beta}$, $\beta \in S^{(R)}$

$$e^{+ix\beta} e^{-ix\alpha} = e^{ix(\beta-\alpha)}$$

$$\Rightarrow \omega = \beta - \alpha \in S^{(R)} - S^{(R)}$$

Key insight: Frequencies arise as *differences of combinatorial eigenvalue sums*.

Why Sums Need Not Match: Example

- U and $U^\dagger$ are generated from the same Hamiltonian
- Hence they share the same eigenvalue set $\sigma(H)$
- But combinatorial sums can differ

Example:

$$\sigma(H) = \{+1, -1\}, \quad R = 2$$

$$\alpha = (+1) + (-1) = 0$$

$$\beta = (+1) + (+1) = 2$$

$$\alpha \neq \beta \quad \Rightarrow \quad \omega = \beta - \alpha = 2$$

Conclusion:

$$\omega \in S^{(R)} - S^{(R)}$$

Key Theoretical Insight:

- Frequency sets K_R grow monotonically:

$$K_1 \subseteq K_2 \subseteq \cdots \subseteq \mathcal{L}$$

- But lattice directions are finite
- $\exists R_{\text{sat}}$ such that:
 - No new frequency *directions* appear for $R \geq R_{\text{sat}}$
 - Only frequency *magnitudes* increase

Theorem: Saturation of Reachable Frequency Directions

Theorem 1 (Saturation of Reachable Frequency Directions).

Let $S \subset \mathbb{Z}^d$ be the finite set of spectral vectors associated with the data-encoding Hamiltonians, and define the integer lattice

$$\mathcal{L} := \text{span}_{\mathbb{Z}}(S).$$

For $R \geq 1$, define:

$$S^{(R)} := \{s_1 + \cdots + s_R \mid s_i \in S\}, \quad K_R := S^{(R)} - S^{(R)}.$$

Then there exists a finite integer R_{sat} such that for all $R \geq R_{\text{sat}}$:

- 1 Every lattice direction in $\mathcal{L}$ appears in K_R
- 2 Increasing R introduces no new frequency directions, only larger magnitudes

Proof Sketch: Geometric Structure

- S is finite $\Rightarrow$ its conical hull

$$\text{cone}(S) = \left\{ \sum_i \alpha_i s_i \mid \alpha_i \geq 0 \right\}$$

is a **polyhedral cone**

- By **Gordan's Lemma**:
 - The integer points in this cone form a finitely generated semigroup
 - There exists a finite **Hilbert basis** $\mathcal{H} = \{h_1, \dots, h_m\}$
- Each h_j is a finite sum of elements of S

Key Insight: Once all basis directions are reachable, no new directions can appear.

Proof Sketch: Saturation

- Each Hilbert basis element admits:

$$h = \sum_{s \in S} n_s s, \quad n_s \in \mathbb{N}$$

with total weight $R(h) = \sum_s n_s$

- Define:

$$R_{\text{sat}} := \max_{h \in \mathcal{H}} R(h)$$

- For all $R \geq R_{\text{sat}}$:
 - Every $h \in \mathcal{H}$ belongs to $S^{(R)}$
 - Any lattice vector $k \in \mathcal{L}$ can be written as:

$$k = u - v, \quad u, v \in S^{(R)}$$

- Hence $k \in K_R$

Conclusion: Reuploading beyond R_{sat} scales frequencies but cannot create new directions.

Intuition Behind Saturation

- QDRs generate Fourier modes on an integer lattice
- Reuploading performs Minkowski sums on spectral vectors
- Finite spectra $\Rightarrow$ finite directional basis
- Depth increases *radius*, not *dimensionality*

Experiments: Function Approximation

- Tested on univariate and multivariate functions
- Dataset: Feynman equations
- Single-qubit QDR with ~ 27 parameters
- Successfully approximates:
 - ReLU, sigmoid, polynomials
 - $e^{\sin(x)}$, step functions

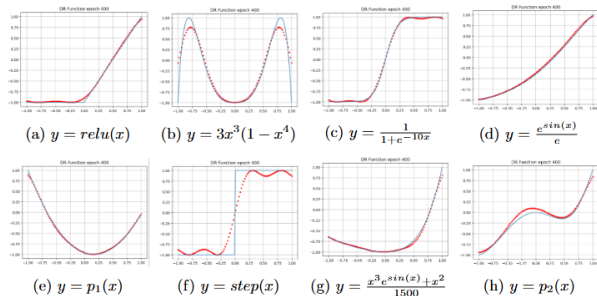


Fig. 1: Plots of various standard univariate functions approximated by a single qubit quantum DRs with 27 parameters and trained for 400 epochs. $y = p_1(x) = \frac{0.01x^4 - 0.1x^3 + 3x^2 - x - 10}{500}$ and $y = p_2(x) = \frac{e^{\sin(x)} \tan(x^3) + (1 - e^{\cos(x)})x^2}{60}$

Effect of Increasing Parameters

- Increasing parameters improves approximation initially
- After a point:
 - Loss stagnates
 - Overfitting observed
- Matches theoretical saturation prediction

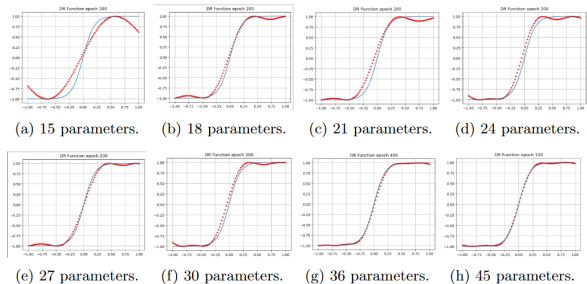


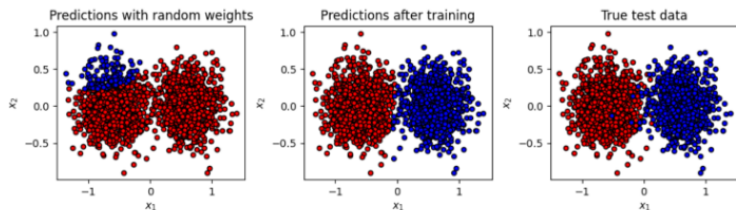
Fig. 2: Plots of tanh equation approximation by QDRs across various parameters

Table 1: Vanilla QDR Performance

Equations 1–14			Equations 15–27		
Equation	Loss	# Params	Equation	Loss	# Params
I.6.2	1.46×10^{-1}	30	I.40.1	1.38×10^{-1}	30
I.6.2b	6.65×10^{-1}	30	I.44.4	2.49×10^{-1}	45
I.9.18	2.07×10^{-1}	30	I.50.26	3.57×10^{-1}	30
I.12.11	1.48×10^{-1}	30	II.2.42	1.68×10^{-1}	45
I.13.12	3.13×10^{-1}	30	II.6.15a	3.89×10^{-1}	45
I.15.3x	3.64×10^0	30	II.11.7	3.66×10^{-1}	45
I.16.6	2.89×10^{-2}	30	II.11.27	2.56×10^{-2}	45
I.18.4	2.83×10^{-1}	30	II.35.18	2.09×10^{-1}	30
I.26.2	1.67×10^{-1}	30	II.36.38	2.36×10^0	30
I.27.6	6.52×10^{-2}	30	II.38.3	1.35×10^{-2}	30
I.29.16	8.03×10^{-1}	30	III.9.52	9.12×10^{-2}	30
I.30.3	1.01×10^{-2}	30	III.10.19	5.34×10^{-1}	30
I.30.5	2.88×10^{-1}	30	III.17.37	3.35×10^{-1}	45
I.37.4	3.36×10^{-1}	30			

QDRs as Classifiers

- XOR, Gaussian blob datasets
- 3-layer QDR, only 9 parameters:
 - XOR: 93% accuracy
 - Gaussian blobs: 98% accuracy
- Complex datasets (spirals):
 - Performance saturates ~85%



(b) QDR on Gaussian Blob dataset with 9 parameters.

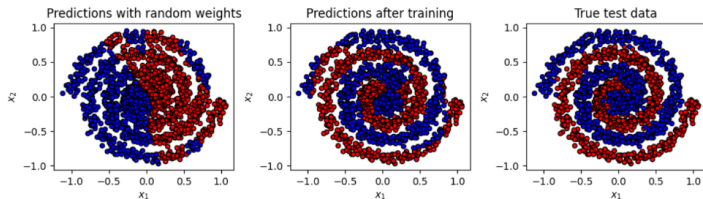


Fig. 3: Spiral dataset classification with 15 layers.

- QDRs compute **band-limited Fourier series**
- Expressivity grows linearly with reuploads
- Saturation limits functional diversity
- Explains empirical plateaus and overfitting
- Future work:
 - Residual QDR architectures
 - Variational action-based QDRs

Thank You

Questions?